ZOO510
Past Final Terms

Objective+
Subjective

Merged by
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Monogastric and Polygastric:

The animal having a simple stomach with only a single compartment is called monogastric.for example: Pig, Dog, Human .The animal having a stomach with more than one compartment is called polygastric.For example: (ruminants) – Cow, Sheep, Goat.

Legs segment in animals:

The jointed appendages are legs segment of animals especially in arthropods. The legs segment of animals that provide travel and protect sensitive organ from damages.

Housefly mouth pars:

Fulcium, clypeus, rostrum, stipes, haustellum,theca, maxillarypalp, labrum, prementum, labellum,pseudotracea, furca and prestomal teeth are mouth parts of housefly.

Vector-Borne Diseases:

diseases : Vectors

1-Dangue: Mosquito 2-Typhus: Rodent 3-Asthma: Cockroach 4-Bubonic Plague: Fleas 5- Skin infection: Louse

Insect life cycle:

1- Egg ----2-larvae----3-pupa----4-adult---

Hemimatabolous and holometabolous:

The gradual or incomplete metamorphosis is called hemimetabolous and complete metamorphous is called holometabolous.

How to control weeds:

Weeds are control by cultural method by using mowing ,grazing , over seeding and improved fertility.

Sources of proteins:

Animals and plants are sources of proteins.

Vector biting Activities:

The nocturnal , diurnal and crepuscular are vector biting activities.

Autogeny and Anautogeny:

A phenomenon in which living organisms are created from non living matter is called autogeny and a strategy in which an adult female insect must eat a particular sort of meal before laying egg in order for her eggs to mature is called anautogeny.

Average milk composition:

The average milk composition is as follows:

Water 87.5% b-Total Milk Solids 12.5% -**Fat 3.8%** - Solids Non Fat (SNF) 8.7% i-**Lactose 4.6%** Protein (Casien) 3.1% -**N Protein 0.2%** -Vitamins & Minerals 0.8%

Plants pathogens:

Pathogens: Fungi – Bacteria – Viruses – Nematodes – Phytoplasmas are the pathogens of plants.

Integrated Pest Management(IPM):

Implementation is done by: 1. Select control strategies 2. Timing 3. Economics 4. Environmental impacts 5. Regulatory restrictions

Why practice IPM:

The IPM practice for to Maintains balanced ecosystems, pesticides alone may be ineffective , promotes a healthy environment ,saves money and maintains a good public image

How many exposures does it take to make your population resistant?

The exposure that make our population resistant are as follows:

Toss your population (pennies) on the ground, and consider all with painted sides showing as exposed to pesticide:

1- All exposed yellow (RR = susceptible) die so remove these pennies.

2– All exposed red (rr = resistant) survive and double; 3-add a red penny for each exposed red penny.

4– For every three exposed blue (mild resistance) add one of each color. (Keep track of any remaining blues

Hygiene and Milk Quality:

A science of the establishment and maintenance of health is called hygiene and white liquid with balance diet is called milk.The hygiene and milk quality is as follows:

The milk without dirt (bacteria or antibiotics, poor milk filters by bacteria or antibiotics, low cream and animal mastitis) is hygiene and milk quality.

Ecological pest management:

The management that focuses on prevent rather than reactive approaches to pest management by increasing biodiversity and creating habitat for natural enemies, land managers can boost natural pest control services and reliance on pesticides in ecology is the ecological pest management.

The natural pest control factors are as follows:

1-Wind 2-Temperature 3-Humidity, rain 4-Rivers, lakes, mountains 5- Pathogens, predators 6-Food supply of the pest

Characteristics of Phylum arhropods:

1-Highly successful and segmented Eucoelomata with exoskeleton (Arthropod Metamerism)
2***Tagmata/Tagmatization*** (segments grouped into body regions)

3-***Molting, Ecdysis (Ecdysozoa)***

4- ***Hemimetabolous (gradual) / holometabolous (complete)***

5- ***Cephalization*** (highly developed sensory organs)

6- ***Open circulatory system (hemolymph / hemocoel)***

7- ***Respiratory system (trachea / book lungs)***

8- ***Malpighian tubules / Coxal glands***

9- **Involved in virtually every kind of parasitic relationship**

What is "Biosecurity"?

Protecting your birds from disease .Preventing or controlling disease transmission by vectors • procedures or measures designed to protect the population against harmful biological or biochemical substances.

What is a vector?

- Something that may transmit a disease • Rodents, birds, insects, shoes, car tires, shared equipment, best friend, pet, feed, water, dust, air…. a vector is an organism that does not cause disease itself but which spreads infection by conveying pathogens from one host to another. Species of mosquito, for example, serve as vectors for the deadly disease Malaria.

How many exposures does it take to make your population resistant?

Expose a population to a pesticide several times with mating in between exposures and see how many exposures it takes for resistant bugs to become the majority.

- **Rules of the Game:**

1. Toss your population (pennies) on the ground, and consider all with painted sides showing as exposed to pesticide
2. All exposed yellow (RR = susceptible) die so remove these pennies.
3. All exposed red (rr = resistant) survive and double; add a red penny for each exposed red penny.
4. For every three exposed blue (mild resistance) add one of each color.

(Keep track of any remaining blues (<3) to ad to those on the next throw) 5.

Keep track of the exposure count and repeat until most pennies are red.

Bioaccumulation:

When a chemical accumulates in animal fat (historical fact) is called bioaccumulation.

Animal by-Products:

The animal by products are as follows:

1-Bones =Button, glue, mineral supplement for livestock feed (Ca)

2--Fat =Chemicals, salves, creams, dressings, lubricants, soaps, food

3- Glands =Medicines, food additives

4- Collagen=Glue, Gelatin

5- Intestinal & Stomach tissue =lunch meats, surgical sutures, strings for musical and sports instruments

6- Fertilizer=Natural

Bubonic Plague:

The plague caused by a bacterium (Yersinia Pestis) and characterized by especially by the formation of buboes is called bubonic plague. This Plague is currently managed monitoring for plague and reducing the number of rodenthosts for fleas.

Treatment of mumbal foot.

a) Soak foot in warm water and Epsom salts. b) disinfect with alcohol. c) If skin is open, drain pus from abscess. d) Flush abscess cavity with hydrogen peroxide to cleanout pus and debris. e) Pack the cavity with antibiotic ointment . f) wrap the foot with gauze and elastic bandage.

Explain lac of insects.

Lac culture is the scientific management of lac insects to obtain a high amount of quality lac. This involves selection and maintenance of host plants, inoculation of host plants with healthy lac insects, collection and processing of lac and protection against enemies. Lac is the resinous secretion of lac insects. Two species of lac insects ***Tachardia lacca and T. chinensis*** are common, of which the former one is predominant in India. India is the highest lac-producing country. Thailand is next in order.

Host plants:

Kusum, Ranjeeni (Khair) and Ber (Plum) trees. The insects live upon plant juice.

Inoculation:

20 to 30 cm long twigs of host plant with old lac crusts are cut and tied with branches of new plants for propagation. The crust contains eggs laid by female insects.

Types of Lac:

1. Kusmi lac:

It grows on Kusum trees. Inoculation is done in January-February and harvested in JuneJuly.

2. Ranjeeni lac:

It grows on trees other than Kusum. Inoculated in October-November and harvested in next May-June.

What is biomagnification.

Bio-magnification: when an organism accumulates residues at higher concentrations than the organisms they consume Some organisms accumulate chemical residues in higher concentrations than those found in their food source. This process is called biomagnification. For example, invertebrates with pesticides in their tissues are eaten by fish, which are then eaten by birds. The birds at the top of the food chain accumulate the highest concentration of pesticide residues. Pesticides that bioaccumulate and biomagnify have been removed from the marketplace. Biomagnification, also known as bioamplification or biological magnification, is the increasing concentration of a substance, such as a toxic chemical, in the tissues tolerant organisms at successively higher levels in a food chain.

Life cycle of insects.

1. Egg

You can find insect eggs everywhere, if you know where to look. Most insects lay their eggs near the food they like to eat. Moths and butterflies lay eggs underneath leaves. Dung beetles lay eggs in balls of animal poop. Many flies lay eggs on meat. Some insects even lay eggs on other animals. Many insects make special cases to protect their eggs. One insect puts foam around its eggs. You may have seen this “cuckoo spit” in

tall grass. Many insects lay eggs in pools, ponds, or streams. Others lay their eggs underground. Rarely, a female insect keeps the eggs inside her body. The eggs hatch, and the young are born alive.

2. Larva .

Most eggs hatch into a worm-like stage called a larva. Caterpillars, grubs, and maggots are larvae (LARvee). They have long bodies and many legs. Almost all larvae are eating machines. A larva can eat more than its own body weight in one day.

Many larvae can destroy crops. Some larvae suck sap from crop roots. Others devour the leaves. Others eat fruits or vegetables. Beetle larvae can weaken and kill trees. Some moth larvae eat wool clothing. On the other hand, the silkworm, a moth larva, makes silk thread.

3. Pupa

Once the larva has grown, it must totally change its shape. And it must protect itself while it does so. Many larvae spin cocoons out of silk or grow hard shells. The pupa does not eat and does not move much. But inside, the insect is very busy changing shape. This change is called metamorphosis (METahMORF-oh-sis).

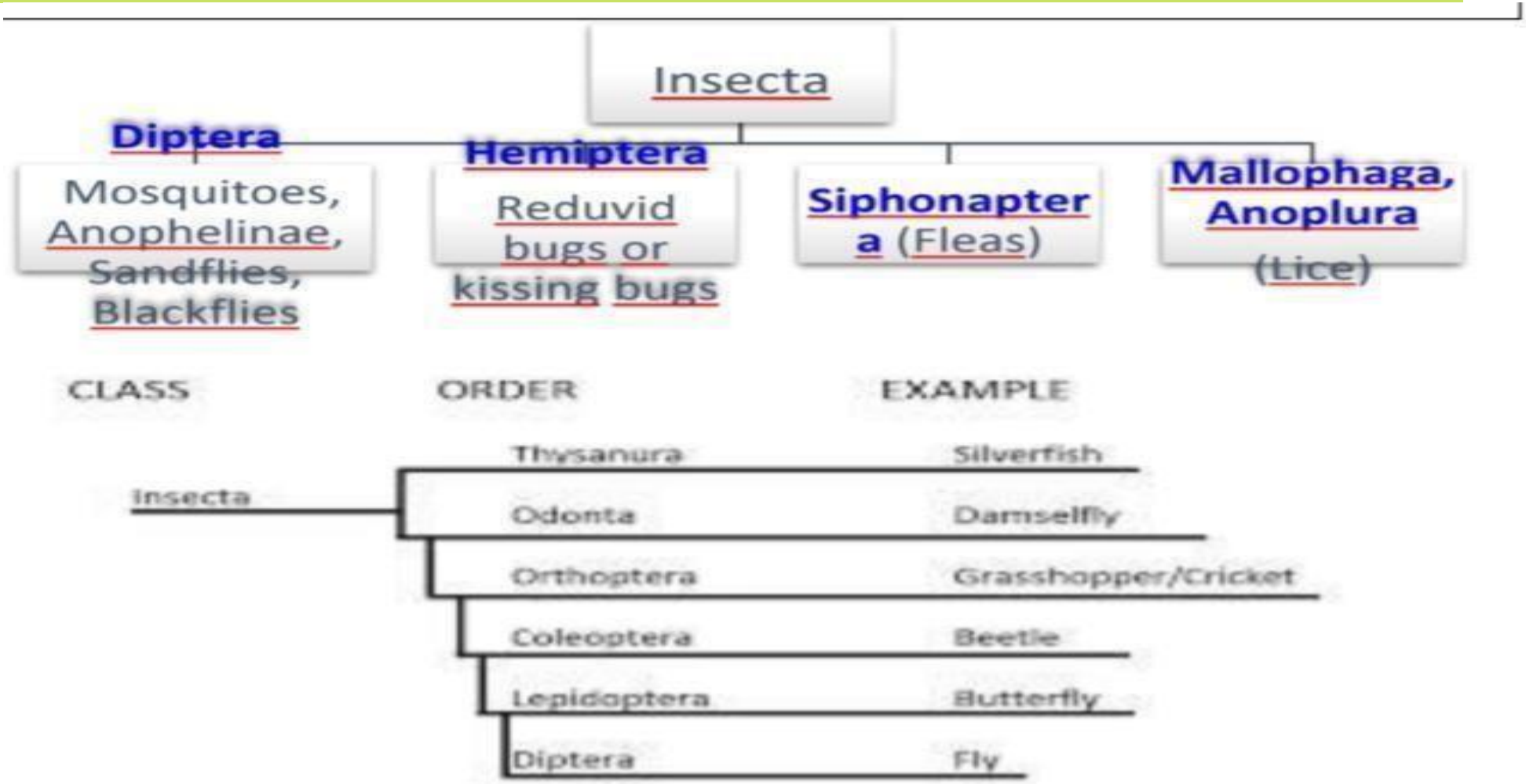
4. Adult

After the metamorphosis is complete, the pupa hatches as an adult. The adult insect has wings, six legs, and three body parts. Many larvae that lived in water will hatch into adults that live on land. Larvae that lived underground will hatch above the soil. Adult insects will find mates, and the females will lay eggs.Sometimes, this is all the adults do. Mayfly adults live for only one day. Most other insects have longer adult lives. They eat, fly, hunt, and work. **5. Nymph**

In the three-stage life cycle, eggs do not hatch as larvae. They hatch as nymphs, which look like small adults. Some nymphs do not have full wings. Others, such as the dragonfly nymph, may live underwater. Nymphs do not turn into pupae. Instead, they shed their skin as they grow. Each time they shed their skin, they look more like an adult. Slowly, the nymph transforms into an adult.

Order of class insecta .

Pest resistance.



The Insecticide Resistance Action Committee (IRAC) definespest resistance as “a heritable change in the

sensitivity of apest population that is reflected in the repeated failure of a product to achieve the expected level of control when used according to the label recommendation for that pest species.

Ecological PestManagement (EPM)

is an approach to increasing the strengths of natural systems to reinforce the natural processes of pest regulation and improve agricultural production. 1. Control based on pest life cycle and ecology

2. Control agent may be an organism or chemical

3. Specific to pest and/or manipulate a part of the ecosystem

4. Emphasizes protection from pest

Monogastric and polygastric .

A monogastric organism has a simple single-chambered stomach, compared with a ruminant organism, like a cow, goat, or sheep, which has a four-chambered complex stomach.

Monogastric (simple stomach)

Pig, Dog, Human

Polygastric (ruminants)

Ruminants are mammals that are able to acquire nutrients from plant-based food by fermenting it in a specialized stomach prior to digestion, principally through microbial actions. The process, which takes place in the front part of the digestive system and therefore is called foregut fermentation, typically requires the fermented ingesta (known as cud) to be regurgitated and chewed again. The process of rechewing the cud to further break down plant matter and stimulate digestion is called rumination Polygastric (ruminants) .. Cow, Sheep, Goat

Common pest control factors.

Natural Pest Controls ***1. Wind 2. Temperature 3. Humidity, rain 4. Rivers, lakes, mountains 5. Pathogens, predators 6. Food supply of the pest 7. Cultural control 8. Control by natural enemies 9. Genetic control 10. Natural chemical control***

Natural controls are out of man's control – Mother nature dishes these out. They are without a doubt the leading factors that manage pest populations. Natural controls include wind, temperature extremes, humidity levels, and rainfall. A cold snap in the spring or a wet spring can largely reduce insect pest numbers for that season. Rivers, lakes, and mountains provide significant barriers to pest movement and expansion. Naturally-occurring predators and pathogens hold down pest populations. The natural food supply of the pest also affects population density.

1st ,2nd ,3rd and 4th generation pesticides.

First-generation pesticides (inorganic)

1. First attempt at chemical technology 2. Included heavy metals such as arsenic, copper and lead.

3. Toxic to humans and agricultural plants. 4. Pests developed resistance.

- **Second-generation pesticides**

1. Organic chemical (organochlorines). 2. Used after WWII (presently in developing countries) 3. Synthesis begins with petroleum ("oil") ✓ Mechanism of actions often unknown. 4. Bioaccumulation & Biomagnification. 5. Toxic to animals (humans) and agricultural plants. 6. Pests developed resistance.

- **Third-generation Pesticides**

1. Organophosphates and carbamates 2. Less persistent in environment (good deal) 3. Acutely potent nerve toxins 4. More lethal in low dose than organochlorines **Fourth generation Pesticides**

1. Endocrine disruptors (hormonal chaos) ✓ Target a critical life cycle stage of insects. 2. Not direct killers per say. 3. Reduce reproduction (fertility) of population.

Source of protein.

Sources of Protein 6 common sources

1. ***Soybean Meal 2. Cotton Seed Meal 3. Fish Meal 4. Tankage 5. Skim Milk 6. Alfalfa***

Difference between hygene quality and milk.

Hygiene and Milk Quality

• Dirty milk (containing dirt, bacteria or antibiotics) can cause serious problems when processing. • Dirty milk is caused by unhygienic milking machines or poor milk filters. • Antibiotics are found in milk as residues from treatments to cows for mastitis. • When milk is found to be not up to standard, it may be rejected by the creamery or bought at a lower price.

Milk composition

Milk used for bottling or drinking must by law contain 3.6% fat and 8.5% SNF. Creameries buy milk in many different ways – sometimes at a flat rate per kg. Sometimes, however, it may be by percentage butterfat or protein. Milk composition varies also amongst different breeds. The milk of the Jersey cow has the highest butterfat and SNF content of any cow. Also the composition of milk varies during milking. The milk at the start may only contain 1% fat, while the milk at the end ("The Stripping") may contain 10% fat.

Bubonic plague .

14th Century Europe: mysterious scourge kills millions .Centuries later it was found that rat fleas became infected with disease-causing bacteria .Fleas sought other warm-blooded hosts (humans) when rat numbers declined .Plague is currently managed monitoring for plague and reducing the number of rodent-hosts for fleas

Do you remember anything from your history lessons about Black Plague in Europe? .During the 14th Century there were Plague outbreaks, millions of people died, and at the time it wasn't understood why. Centuries later, it was discovered that Plague was a bacterial disease and it was spread by fleas found in rodent populations. Besides biting rats, infected fleas would bite humans and other mammals. Bubonic plague is still present in some rodent populations today, even here in the United States. But with rodent management and plague monitoring, only a few people a year are infected with plague.

Bio accumulation

Bioaccumulation: when a chemical accumulates in animal fat (historical fact) .Bioaccumulation is the gradual accumulation of substances, such as pesticides, or other chemicals in an organism.

Bioaccumulation occurs when an organism absorbs a substance at a rate faster than that at which the substance is lost by catabolism and excretion.

Mode of action of pesticide and pest resistance.

Mode of action: how they work to control the pest

– **Systemic pesticides** are absorbed through tissues and transported elsewhere where the pest encounters it through feeding • Used on plants or livestock

– **Contact pesticides** must come in direct contact with the target pest .Pesticides vary in how they work to actually control the pest. We call this the mode of action –how they kill or control the pest. Some pesticides are systemic and actually move through animal or plant tissues and the pest encounters it when feeding on the animal or plant. Other pesticides work by contact action like battery acid on your pants. You must make sure that you apply the material so that it comes into direct contact with the pest.

Selectivity:

what range of pests they affect

a)Non-selective – kills all related pests – for example some herbicides kill all green plant that gets a sufficient dose

b)Selective – kills only certain weeds, insects, plant pathogens – for example other herbicides only kill broadleaf weeds not grasses .Pesticides also vary in their selectivity or how many different pest species they target. Some affect all related organisms equally and are called non-selective. A good example is glyphosate, the active ingredient in Roundup Original. Glyphosate is non-selective when applied to green

plants. ANY green plant that gets a sufficient dose will die whether it's a broadleaf or a grass plant. In contrast, a selective pesticide only kills a small group of related organisms and does not harm others. The herbicide 2,4-D is a good example. It only kills broadleaf plants and does not harm grass plants when applied according to label directions.

PesT Resistance

The ability of insect pests to resist against pesticides .: how long they remain active in the environment ***Residual pesticides*** remain active for weeks, months, years – for example herbicides used around road guard rails

Non-residual inactivated immediately or within a few days – for example some herbicides do not remain active in the soil once applied .Persistence is the term used for how long a pesticide remains present in the environment. DDT, which is no longer used, was a very persistent insecticide that's still found in the environment today. Many of the new products break down in a matter of hours or a few weeks. There are some persistent materials on the market today; we call them residual insecticides and herbicides and use them where we want long-term control. Applications of residual herbicides are made to rights-of-ways so repeated applications are not required throughout the year. Non-residual materials are also available, they break down very quickly.

1. At an age of 10-12 weeks cocks are separated from …_____ hens
2. ….. is the first limiting AA for egg production in laying hens_____ methionine
3. Geese will start to eat pasture when they are only …old _____ few days
4. The color of egg yolk depend upon the presence of _____carotenoid
5. Very hot weather causes ..quality egg shells _____ poor

6. formulating poultry diets …. is the most common grain _____ Corn
7. In turkeys sticky droppings adhere to the feet & cause dermatitis_____ Footpad dermatitis
8. In feeding of broilers starter ration contains ….. % protein. _____22-24
9. inflammation of tongue & mouth cavity (black tongue) caused by _____ niacin
10. Deficiency of niacin, biotin, Vit E & zinc causes _____ Enlargment of hock joint
11. For egg production.....is first limiting AAs_____ Methionine
12. Chickens bred for meat production….. grow and achieve adult stage much early_____
13. _____rapidly
14. Period after ….. of age hens attained mature body weight. _____ 42w
15. Drying and exposure to..... are very effective in killing many disease causing
16. organisms._____ sunlight
17. is the most common grain used in formulating poultry feeds _____ corn
18. Inorganic…. is supplied by bone meal, di-calcium phosphate, and rock phosphate in poultry
19. feeds_____ phosphate

1- ….is farming of aquatic organisms in natural or controlled marine or Fresh water environment.

_

20. _____Aquaculture
21. 2- RAS stands for._____Recirculating aquaculture system
22. 3- Induction air into water is called .._____Aeration
23. 4- Transformation from soft, dusty mixture to compact form is._____Pelleting
24. fungi are._____multicellular
25. Most common feeding method for fishes is._____Hand feeding
26. Bombyx mori produces.._____Silk
27. bombyx mori produces ...-.. percent of the worlds raw silk _____90%.
28. In..... Lac insect is known to have two distinct strains: kusumi and rangeeni. _____india
29. 10-The life cycle of lac insect takes about..... Months _____ 6
30. 11-Mature male chicken is called. _____Roaster
31. 12-The lac which grows on non-kusum plants is called as.._____Ranjeem lac 32. 13-India alone accounts for about ... percent of global lac production. _____ 70%
33. 14- QMP stands for.._____Queen mandibular pheromone.
34. 15-.....is a farming of chicken, turkey, duck and goose. _____ poultry

35. 16-Raceways are considered.._____Flow through systems
36. 17- .....is a silkworm disease _____Pebrine
37. 18-Aghani and baisakhi strain are the main crops contributing about..... Percent of lac production.
38. _____90%
39. 19-Lac cultivation is introduced into..... From India. _____ Thailand
40. 20-a group of chicken is called ……_____flock 41. 21Trained staff called …….'catch the birds and put them in special containers called …….
42. _____ catchers' .modules.
43. 22-life of worker bee is _____28 to 35 days.
44. 23-Carnivorous fish consume foods with …..of protein ._____ ~50%
45. 24-The place where a beekeeper keeps his bees is called an _____apiary or a bee yard.
46. 25-Typically ….. pipe is sufficient for ponds less than 10 acres . _____6 inch
47. 26-Whirling disease Caused by. _____Myxobolus cerebralis
48. 27-Mulberry culture spread in India by 140BC from _____China
49. 28-The common parasites of lac insect are known as_____ "Chalcid."

50. 29-A virus is a _____microscopic organism
51. 30-American Cancer Association Regular fish consumption decreases chances of colon
52. cancer_____50%
53. -Unsaturated FA = _____double bonds
54. -James I introduced sericulture to ….He also took sericulture to.. _____ England America.
55. in poultry feeding some medicines are administered in the....._____ water
56. Deficiency of in the ration of laying hens cause goiter,_____ Iodine
57. Sources of…. in poultry diets are ground oyster shell, limestone, bone meal, and di-calcium
58. phosphate,_____ Ca
59. if a commercial feed for ducks & geese is not available… feed may be used ____ Chicken
60. Laying hens (2kg wt.) requires..... Kcal energy for their maintenance_____ 220
61. Animal protein sources are more variable in their amino acids than….protein _____ plant
62. Young chick requires more niacin than….. due to less bacterial activity _____ adult
63. is the most common grain used in formulating poultry feeds _____ Corn
64. -Female lac insect is …than male. _____larger

65. (blood poisoning due to bacteria. _____ Septicemia
66. -a parasite which lives inside the host is an _____endo parasite.
67. Spring Viremia of Carp A …..caused by Rhabdovirus carpio. _____ viral infection
68. To produce 1 kg of lac resin, around …insects lose their life. _____ 300,000
69. ….caused by fish disease. _____ Aeromonas hydrophila
70. Feeds should be stored in a ….area . _____ dry, cool and well-ventilated
71. Male Lac-insect: The body is typically divided into head, …and abdomen. _____ thorax were used as one of the first forms of aquaculture. _____Ponds
72. _____ is the largest aquaculture producing country in the world. _____China 73. _____ were used as one of the first forms of aquaculture. _____Ponds
74. CO can be removed by trickling filter by keeping gas/liquid ratio of_____. _____ 5:1
75. Which one of the following is the Vitamin is water soluble? _____Vitamin B
76. Whirling disease is caused by the_____. _____Protozoan
77. Which one of the following is included in silkworm disease? _____Perrine
78. In_____ Lac insect is known to have two distinct strains, Kusuma and Range Eni. _____ India
79. The common parasites of lac insect are known as_____. _____Chalcid

80. Chicken, turkey, duck and goose are all types of birds and they are reared for meat is
81. called_____. _____Poultry
82. A group of chickens is called a_____. _____Flock
83. A group of baby chickens is called a_____. _____Peep
84. _____is first limiting AAs for egg production. _____ Methionine
85. If a commercial feed for ducks & geese is not available ___ feed may be used. ____Chicken
86. Which one of the following is not an infectious agent? _____ Physical 87. Fowl pox vaccine at _____ weeks of age. _____ 4-6
88. Which one of the following is livestock selection based?
89. _____Based on performance of ancestors
90. The total amount of feed an animal has in a _____ period. _____ 24 hours
91. Which is the by-product of Corn having residue? _____Removal of Starch
92. _____ high stocking rate of animals are put into a pasture for a limited period. 93. _____Strip Grazing
94. Which statement is true about the Biological Control?

95. _____Avoid broad-spectrum insecticides
96. Humidity, temperature, air movement, water and light are the _____ factor. __ Physical
97. Commercial feeds in …form available for ducks & geese __ mash, pelleted or crumbles
98. Performance of livestock is based or efficiency of production which is the ability to utilize feed
99. efficiently, _____in producing meat, milk, wool or power.
100. …Should be fairly fine-chopped to ensure packing _____ Corn Stover (Stalklage)
101. ….Major Pest Categories . _____ Four
102. Complex learning (intelligence) . _____ genetics
103. The color of egg yolk depend upon the presence of carotenoid pigment (…) in the ration
104. _____ xanthophylls
105. Fowl pox vaccine at 4-6 weeks of age ➢ Second dose at 12-14 weeks of age.
106. Evaluating Hay Quality involved _____Odor and Condition
107. rape, turnips, or summer grasses example of _____Strip Grazing
108. Ticks sense …warmth and movement from animals _____ CO2,
109. Intensive pesticide use kills susceptible …in a population, leaving some resistant ones to reproduce

110. _____ pests
111. …aphids per wheat plant = no problem - no action _____6
112. Making good Silage moisture range target . _____ 50-70%
113. ……: def. Of niacin, biotin, vit.E & zinc._____ Enlargement of hock joints Cause
114. …is needed of egg production & hatchability. _____Maganese
115. ….is first limiting AAs for egg production. _____ Methionine
116. When …is found to be not up to standard, it may be rejected by the creamery._____ miks 117. Mesogenic strain (R2B) for the treatment of ranikhet disease useful at the age of 118. _____6-8 weeks of age.
119. Which one is the Lab Animals -_____ Mice,
120. Good Biosecurity Practices should Reduce exposure form. _____All the above.
121. Staphylococcus bacteria causes. _____BUMBLE FOOT
122. The milk of the …has the highest butterfat and SNF content of any co_____ Jersey cow
123. Fat Soluble Vitamin _____A,
124. NH3 forms are dependent on _____ Both
125. Commonly Cultured bait Fish species_____ ALL

126. Stripped Bass is an example of…. Commonly cultured fish specie _____ Food fish
127. industrialized Aquaculture are _____ Intensive Aquaculture
128. Which of the following is not a type of aquaculture systems?_____Fresh water culture
129. In levee pond construction, maximum slope of levee should be._____4:1
130. 5-7 acre watershed per surface acre of water is disadvantage of _____ surface
131. is a common method of feeding fish_____ Hand Feeding
132. Commonly cultured food fish species include, _____Catfish
133. Rainbow trout feed conversion _____None
134. First fish hatchery was in Oregon_____1877
135. Construction of pands include? _____All
136. Which of the following is not included in species selection? _____Yield control
137. According to american cancer association regular fish consumption decreases chances of
138. cancer by _____50%
139. Farming implies some form of intervention in the rearing process to enhance production, such
140. as _____ All of the above

141. …..impacts the rates of chemical and bialogical process at the most fundamental level
142. ._____Temperature
143. Economic zoology deals with animal world that is associated with the. ____All of the above
144. Omege-3 fatty acids decreases occurrence of _____Heart disease
145. ….is the old name marine or brackish water._____Mariculture
146. Aquaculture encompass farming of aquatic organisms. Including ,_____All of the above
147. in pond design criteria. Levees with slopes 1:2 for _____High clay
148. causes disease in fish _____ Aeromonas hydrophobia
149. Species selection include? _____All
150. If ponds must be drained to harvest (shrimp) then use one per acre of water_____10 inch
151. in pond design criteria Slope to harvest basin should be. _____0.5-1.0%
152. A process in which NH3 is converted to NO3 is known as _____Nitrification
153. inhabiting macro and micro life Aquarium trade? _____ live rock
154. Low oxygen is disadvantage of _____Wells
155. Which of the following specie(s) can be cultured in ponds? _____All

156. First book on fish farming _____475 B.C
157. Abalone are commonly cultured. _____Mollusks
158. induction of pure oxygen is called _____Oxygenation
159. High density aquaculture are _____Intensive aquaculture
160. Who wrote first book on fish farming? _____None
161. Beef cattle on feedlot feed conversion ratio is. _____8:1
162. Watershed ponds Requires acres of watershed per acre of water _____5-7
163. Compost is fertilizer. _____Organic
164. In aquaculture there are. primary sources of nitrogenous wastes _____4
165. Minimal control and lower density are_ _____extensive aquaculture
166. In pond design criteria. Once formed, levees are sprigged with grass to erosion.__ __Reduce
167. To produce 1 kg of lac resin, around insects lose their life_____300,000
168. Stocking density is fish ….per acre._____800-1000
169. Dissolved oxygen and water temperature usually very over a hour cycle._____24 170. 3 species culture include. _____All
171. is a disease of silkworms_____Pebrine

172. Which of the following filters are used with various mesh screens. _____Screen filters
173. in outgoing bio filter water not less than, ppm._____2
174. Which of the following is considered as disadvantage of surface? _____Contaminates
175. In fish tank O2 should not be less than ppm._____5
176. Which of the following describes the bulk of high-density production systems where risk and
177. economics must be carefully balanced to achieve profitability _____Mesotrophic
178. In pond design criteria Heights are determined by pond. _____Both
179. Mechanical filtration include? _____Both a&b
180. is/are highly soluble in water. _____All
181. Semi closed system water reuse rates are_____50%
182. Construction of ponds include? _____All
183. are used for extraction of medicines. _____All
184. An independent for each pond if filled by tube well. _____Electrical source
185. Which of the following parameter is/are considered to design aquaculture _____All
186. levee pond construction, minimum slope of levee should be._____3:1

187. In ponds construction.number/size of gates depends on of harvest required _____Speed
188. Levee pond width should be._____20 ft
189. Carp were farmed in China as early as _____2500 BC
190. needed for egg production & hatchability_ _____Mn
191. Commonly cultured ornamental fish species are. _____Backyard Bands

ZOO510-QUIZ 2, FALL21

Solved by Muhammad Kashif

Helpzone Group of Zoologists

0306-4963933

ZOO510 - Economic Zoology (Quiz 2)

Question # 1 of 10 (Start time: 12:03:17 AM, 21 February 2022)

In bumble foot common causes of injury is.....

Select the correct option

○	stones
○	wood
●	wire floors
○	thorns

Solved by Muhammad Kashif

Question # 2 of 10 (Start time: 12:04:37 AM, 21 February 2022)

In poultry Bacillary white diarrhea is treated with .....

Select the correct option

Solved by Muhammad Kashif

○	Cobalamin
●	Enrofloxacin
○	Niacin
○	Folic acid

ZOO510 - Economic Zoology (Quiz 2)

Question # 3 of 10 (Start time: 12:05:38 AM, 21 February 2022)

The cow repeats its oestrous cycle after of a cow is .....and it lasts for 18 hours on the average.

Select the correct option

- 31 days
- 21 days
- 51 days
- 41 days

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 4 of 10 (Start time: 12:06:44 AM, 21 February 2022)

A female calve (destined for dairying) approximately weighs ..... at birth.

Select the correct option

●	40kg
○	50kg
○	60kg
○	70kg

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 5 of 10 (Start time: 12:07:20 AM, 21 February 2022)

In nutritional diseases appearance of curled toe paralysis is due to the deficiency of .....

Select the correct option

- riboflavin
- folic acid
- niacin
- thiamine

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 6 of 10 (Start time: 12:08:07 AM, 21 February 2022)

Good Biosecurity practices should reduce the exposure from

Select the correct option

Solved by Muhammad Kashif

○	insects
○	rodents
●	all of above
○	birds

ZOO510 - Economic Zoology (Quiz 2)

Question # 7 of 10 (Start time: 12:08:52 AM, 21 February 2022)

In poultry Bacillary white diarrhea clinical signs is .....

Select the correct option

- baldness
- sluggish
- Weakness
- itching

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 8 of 10 (Start time: 12:09:44 AM, 21 February 2022)

Livstock raising is still ----------despite numerous advancements in science.

Select the correct option

Solved by Muhammad Kashif

- (●) Traditional
- ( ) advance
- ( ) useful
- ( ) bore

ZOO510 - Economic Zoology (Quiz 2)

Question # 9 of 10 (Start time: 12:11:01 AM, 21 February 2022)

Disease is the any condition that results in the deviation from.....

Select the correct option

Solved by Muhammad Kashif

- environment
- species
- drinking
- normal function

ZOO510 - Economic Zoology (Quiz 2)

Question # 10 of 10 (Start time: 12:11:46 AM, 21 February 2022)

----------is the most common sign of illness in poultry farming.

Select the correct option

- (●) Poor appetite
- ( ) vomit
- ( ) cough
- ( ) sleeping

Solved by Muhammad Kashif

Question # 1 of 10 (Start time: 12:23:12 AM, 21 February 2022)

In poultry Bacillary white diarrhea is treated with .....

Select the correct option

○	Cobalamin
●	Enrofloxacin
○	Niacin
○	Folic acid

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 2 of 10 (Start time: 12:23:42 AM, 21 February 2022)

Good Biosecurity practices should reduce the exposure from

Select the correct option

- (●) all of above
- ( ) rodents
- ( ) birds
- ( ) insects

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 3 of 10 (Start time: 12:23:57 AM, 21 February 2022)

Disease is the any condition that results in the deviation from.....

Select the correct option

- drinking
- normal function
- species
- environment

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 4 of 10 (Start time: 12:24:12 AM, 21 February 2022)

For the maintenance of healthy flock, always try to provide .....as per requirements of the animals.

Solved by Muhammad Kashif

Select the correct option

- (●) all of these
- ( ) Clean water
- ( ) Fresh feed
- ( ) dry litter

ZOO510 - Economic Zoology (Quiz 2)

Question # 5 of 10 (Start time: 12:25:44 AM, 21 February 2022)

Animals provide us as food.

Solved by Muhammad Kashif

Select the correct option

- (●) all of these
- ( ) meat
- ( ) egg
- ( ) milk

ZOO510 - Economic Zoology (Quiz 2)

Question # 6 of 10 (Start time: 12:26:11 AM, 21 February 2022)

In poultry Bacillary white diarrhea clinical signs is .....

Select the correct option

Solved by Muhammad Kashif

○	itching
○	sluggish
○	baldness
●	Weakness

ZOO510 - Economic Zoology (Quiz 2)

Question # 7 of 10 (Start time: 12:26:46 AM, 21 February 2022)

Econimically important species of animals which can be raised in captivity are .....

Select the correct option

Solved by Muhammad Kashif

- ○ cattle
- ○ horses
- ○ sheep
- ● all of these

ZOO510 - Economic Zoology (Quiz 2)

Question # 8 of 10 (Start time: 12:27:53 AM, 21 February 2022)

Livstock raising is still ----------despite numerous advancements in science.

Select the correct option

- bore
- advance
- useful
- Traditional

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 9 of 10 (Start time: 12:28:06 AM, 21 February 2022)

A female calve (destined for dairying) approximately weighs ..... at birth.

Select the correct option

- 70kg
- 50kg
- 60kg
- 40kg

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 10 of 10 (Start time: 12:28:22 AM, 21 February 2022)

----------is the most common sign of illness in poultry farming.

Select the correct option

- vomit
- sleeping
- Poor appetite
- cough

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 1 of 10 (Start time: 12:34:12 AM, 21 February 2022)

----------is the most common sign of illness in poultry farming.

Select the correct option

- cough
- sleeping
- vomit
- Poor appetite

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 4 of 10 (Start time: 12:34:39 AM, 21 February 2022)

Check birds periodically for lice, mites, and .....

Select the correct option

Solved by Muhammad Kashif

- (●) worms
- ( ) rat
- ( ) predator
- ( ) prey

ZOO510 - Economic Zoology (Quiz 2)

Question # 6 of 10 (Start time: 12:35:18 AM, 21 February 2022)

Staphylococcus bacteria causes

Select the correct option

Solved by Muhammad Kashif

- (●) BUMBLE FOOT
- ( ) Ranikhet disease
- ( ) Cryptosporidiosis
- ( ) Coccidiosis

ZOO510 - Economic Zoology (Quiz 2)

Question # 8 of 10 (Start time: 12:35:59 AM, 21 February 2022)

Among other uses some animals are used as beast of burden and serve the purpose of ----

Select the correct option

Solved by Muhammad Kashif

- leather
- pets
- Transportation
- wool

ZOO510 - Economic Zoology (Quiz 2)

Question # 10 of 10 (Start time: 12:37:40 AM, 21 February 2022)

Treatment of bumble foot is to disinfect the affected place with .....

Solved by Muhammad Kashif

Select the correct option

- milk
- Alcohol
- antibiotic
- water

ZOO510 - Economic Zoology (Quiz 2)

Question # 4 of 10 (Start time: 12:42:58 AM, 21 February 2022)

Cow suckles it calves and usually produces milk for up to..... years.

Select the correct option

- (●) 2
- ( ) 3
- ( ) 4
- ( ) 5

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 7 of 10 (Start time: 12:43:56 AM, 21 February 2022)

In Etiology Non-infectious agents is/are.....

Select the correct option

- (●) all of these
- ( ) Physical
- ( ) Chemical
- ( ) Toxins

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 2)

Question # 5 of 10 (Start time: 12:48:10 AM, 21 February 2022)

Mesogenic strain (R2B) FOR THE TREATMENT OF RANIKHET DISEASE USEFULL AT THE AGE OF

Select the correct option

Solved by Muhammad Kashif

- ○ 12-18 weeks of age
- ○ 16-20 weeks of age
- ○ 10-18 weeks of age
- ● 6-8 weeks of age

ZOO510 - Economic Zoology (Quiz 2)

Question # 7 of 10 (Start time: 12:49:14 AM, 21 February 2022)

which one is the lab animal

Select the correct option

●	Mice
○	Deer
○	Cat
○	Cattle

Solved by Muhammad Kashif

QUIZ 3
FALL 21

ZOO510

VU HELPZONE

GROUO OF
ZOOLOGISTA



MARCH 2
DUE DATE
2022

MUHAMMAD KASHIF
03064963933

ZOO510 - Economic Zoology (Quiz 3) Quiz Start Time: 02:24 AM, 01

Question # 10 of 10 (Start time: 02:25:24 AM, 01 March 2022)

Integrated pest management relies on....., measurement, assessment, and knowledge of pest.

Solved by Muhammad Kashif

Select the correct option

- control
- practice
- identification
- limitation

ZOO510 - Economic Zoology (Quiz 3) Quiz Start Time: 02:2

Question # 7 of 10 (Start time: 02:24:58 AM, 01 March 2022)

.....generation pesticides are synthesized with the incorporation of petroleum ("oil").

Select the correct option

- first
- second
- third
- fourth

Solved by Muhammad Kashif

Click to Save Answer & Move

ZOO510 - Economic Zoology (Quiz 3)

Question # 1 of 10 (Start time: 02:20:48 AM, 01 March 2022)

Alteration in the conditions or pest behaviors is called .....

Select the correct option

- sanitation
- fertilizer
- irrigation
- fertilization

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 3)

Question # 7 of 10 (Start time: 02:14:43 AM, 01 March 2022)

Manipulation of pests hormones or pheromones disrupt the .....of pest.

Select the correct option

- weeds
- pest
- insect
- life cycle

Solved by Muhammad Kashif

Click to

Question # 5 of 10 (Start time: 02:13:55 AM, 01 March 2022)

.....generation pesticides are synthesized with the incorporation of petroleum ("oil").

Select the correct option

- first
- fourth
- second
- third

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 3)

Question # 8 of 10 (Start time: 02:08:02 AM, 01 March 2022)

Soil conditions are disrupted to eradicate weeds and .....

Select the correct option

- fungi
- air
- Insects
- pest

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 3)

Question # 10 of 10 (Start time: 02:05:14 AM, 01 March 2022)

Monitor airports, ocean ports and .....to control a pandemic.

Select the correct option

- (●) borders
- ( ) lakes
- ( ) houses
- ( ) fields

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 3)

Question # 9 of 10 (Start time: 02:04:48 AM, 01 March 2022)

Identify the .....and know its biology.

Select the correct option

- plant
- animal
- pest
- insect

Solved by Muhammad Kashif

Cli

ZOO510 - Economic Zoology (Quiz 3)

Question # 8 of 10 (Start time: 02:03:54 AM, 01 March 2022)

Any material that is applied to kill the pests is called .....

Select the correct option

○	chemical
●	pesticide
○	fungicide
○	herbicide

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 3)

Question # 3 of 10 (Start time: 02:02:50 AM, 01 March 2022)

Gene may have various forms which are called .....

Select the correct option

○	fittness
●	Alleles
○	gene flow
○	genetic drift

Solved by Muhammad Kashif

Question # 10 of 10 (Start time: 01:54:34 AM, 01 March 2022)

Release of natural enemies their subsequent establishment may .....infestation levels.

Solved by Muhammad Kashif

Select the correct option

○	fluctuate
○	constant
●	reduce
○	increase

ZOO510 - Economic Zoology (Quiz 3)

Question # 9 of 10 (Start time: 01:53:47 AM, 01 March 2022)

Populations will have different proportions of dominant and recessive traits depending on .....in the environment.

Select the correct option

- selective pressures
- fitness
- gene flow
- genetic drift

Solved by Muhammad Kashif

Click to S

ZOO510 - Economic Zoology (Quiz 3)

Question # 7 of 10 (Start time: 01:51:26 AM, 01 March 2022)

In genetic control sterile ..... are released into pest population.

Select the correct option

- young ones
- males
- females
- pest

Solved by Muhammad Kashif

ZOO510 - Economic Zoology (Quiz 3)

Question # 6 of 10 (Start time: 01:50:26 AM, 01 March 2022)

The..... is the combined effect of both chemicals are typically fat soluble and persistent.

Solved by Muhammad Kashif

Select the correct option

- (●) bioaccumulation
- ( ) bio-leaching
- ( ) bio-concentration
- ( ) bio-filtration

ZOO510 - Economic Zoology (Quiz 3)

Question # 5 of 10 (Start time: 01:49:18 AM, 01 March 2022)

Fourth-generation pesticides target a critical stage of..... in insects.

Solved by Muhammad Kashif

Select the correct option

- (●) life cycle
- ( ) adult
- ( ) pupa
- ( ) larvae

Click

ZOO510 - Economic Zoology (Quiz 3)

Question # 4 of 10 (Start time: 01:49:03 AM, 01 March 2022)

Third-generation pesticides are organophosphates and .....

Solved by Muhammad Kashif

Select the correct option

- silicates
- nitrates
- carbamates
- sulfates

Click to S

ZOO510 - Economic Zoology (Quiz 3)

Question # 3 of 10 (Start time: 01:48:43 AM, 01 March 2022)

Pests are not native to any specific.....

Select the correct option

●	all of these
○	area
○	house
○	field

Solved by Muhammad Kashif

Click

ZOO510 - Economic Zoology (Quiz 3)

Question # 2 of 10 (Start time: 01:48:30 AM, 01 March 2022)

Integrated pest management promotes a .....environment.

Solved by Muhammad Kashif

Select the correct option

- unhealthy
- polluted
- healthy
- fresh

ZOO510 - Economic Zoology (Quiz 3)

Question # 1 of 10 (Start time: 01:47:36 AM, 01 March 2022)

Set IPM goals to prevent, suppress and .....the pests.

Select the correct option

○	flourish
●	eradicate
○	relief
○	control

Solved by Muhammad Kashif

Click t

ZOO510 - Economic Zoology (Quiz 3)

Question # 10 of 10 (Start time: 01:42:13 AM, 01 March 2022)

The.....prevents pests from entry to an area or movement from infested areas.

Solved by Muhammad Kashif

Select the correct option

- restriction
- quarantine
- boundary
- fencing

ZOO510 - Economic Zoology (Quiz 3) Quiz Start Time: 01:35

Question # 9 of 10 (Start time: 01:41:37 AM, 01 March 2022)

Populations will have different proportions of dominant and recessive traits depending on .....in the environment.

Select the correct option

- genetic drift
- selective pressures
- fitness
- gene flow

Solved by Muhammad Kashif

Click to Save Answer & Move t

ZOO510 - Economic Zoology (Quiz 3)

Question # 8 of 10 (Start time: 01:40:55 AM, 01 March 2022)

As..... destroys infected crop residues or infected ornamental plant materials.

Solved by Muhammad Kashif

Select the correct option

- (●) sanitation
- ( ) fertilizer
- ( ) irrigation
- ( ) flood

ZOO510 - Economic Zoology (Quiz 3)

Question # 5 of 10 (Start time: 01:39:20 AM, 01 March 2022)

Bio-concentration is a movement against a concentration gradient typically .....

Select the correct option

○	none of these
●	fat soluble
○	lipid soluble
○	water soluble

Solved by Muhammad Kashif

Click

ZOO510 - Economic Zoology (Quiz 3)

Question # 4 of 10 (Start time: 01:38:11 AM, 01 March 2022)

Pests developed .....system against used pesticides.

Solved by Muhammad Kashif

Select the correct option

- (●) resistance
- ( ) respiration
- ( ) olfaction
- ( ) nervous

ZOO510 - Economic Zoology (Quiz 3)

Question # 3 of 10 (Start time: 01:37:26 AM, 01 March 2022)

Good quality plants or animals are bred to resist the attack of .....

Select the correct option

- insect
- pests
- animals
- weeds

Solved by Muhammad Kashif

Click to S

ZOO510 - Economic Zoology (Quiz 3)

Question # 1 of 10 (Start time: 01:35:03 AM, 01 March 2022)

The..... is the combined effect of both chemicals are typically fat soluble and persistent.

Select the correct option

- bio-filtration
- bioaccumulation
- bio-concentration
- bio-leaching

Solved by Muhammad Kashif